PROJECT DOCUMENTATION

Solution Architecture

Architecture of the F1 dataset, Tableau analytics and Bootstrap website

Date	19 September 2026	**Team ID**	
Project Name	F1 Racing Analytics	**Maximum Marks**	4 Marks

Architecture overview

The solution follows a lightweight, read-only analytics architecture. Structured F1 source files are connected to Tableau, analysed using Tableau fields/calculations, published through Tableau Public, and embedded in the website for visitors.

Logical data flow

Layer	Role in this project	Technology / Component
1. Data Sources	Structured F1 records provided as CSV/text files.	CSV / text files
2. Data Ingestion / Connection	Connect source files to the Tableau workbook; combine compatible files where required.	Tableau data connections / union where applicable
3. Data Preparation / Processing	Use available fields, clean/organise data as required, and apply filters and aggregations.	Tableau Desktop data pane and calculations
4. Analytics / BI Layer	Calculate points totals, rankings and counts used by the visualisations.	Tableau measures, calculated fields, filters and Top N control
5. Reporting / Visualization	Create and publish dashboard sheets and the driver leaderboard story.	Tableau Public
6. Website / Presentation Layer	Provide Home, Overview, Dashboard and Story navigation and embed the published views.	HTML, CSS, Bootstrap, JavaScript
7. Business Users	Explore F1 performance patterns and compare drivers, teams and race-related metrics.	Web browser with internet access

Text-based architecture diagram

F1 CSV / text files → Tableau data connection → Tableau preparation & calculations → Tableau dashboard and story published on Tableau Public → Bootstrap website embeds → Visitor explores charts and filters.

Component Description Table

Component Name	Description / Role in Architecture	Technology Used
Source dataset	Provides the records analysed by the project.	CSV / text dataset files

Component Name	Description / Role in Architecture	Technology Used
Analytics workbook	Connects data and defines views, measures, filters and Top N interaction.	Tableau Desktop / Tableau workbook
Published visualisations	Hosts shareable dashboard and driver leaderboard story.	Tableau Public
Website interface	Presents project introduction and navigation; hosts embedded Tableau views.	HTML, CSS, Bootstrap, JavaScript

Dependencies and boundaries

The dashboard/story embeds require internet access and the published Tableau views to remain available. Interactivity is provided by the embedded Tableau views and their configured controls. No separate SQL database, Python service, authentication system or API backend is assumed in this architecture.